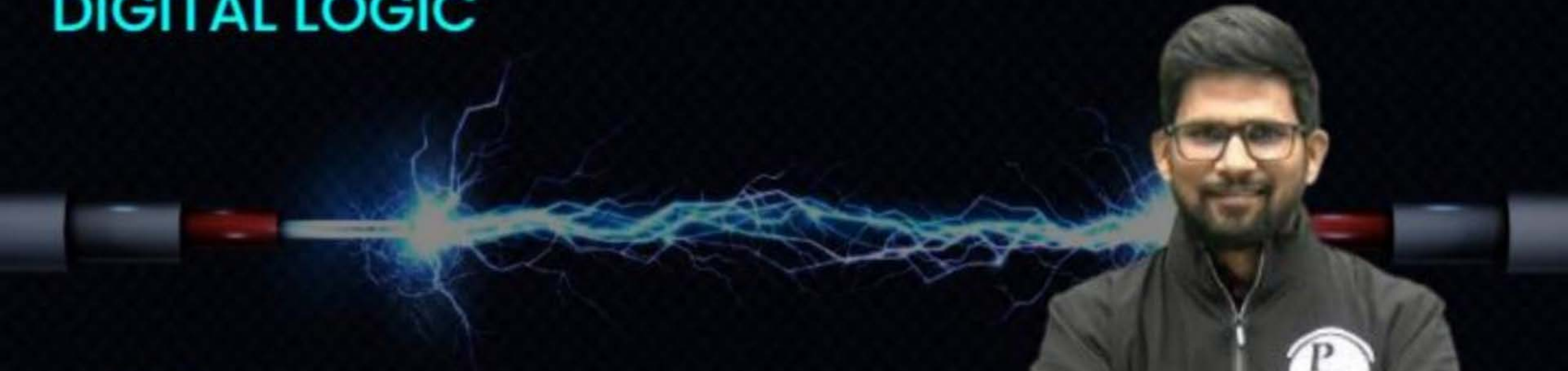
COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No. 13

Combinational Circuit



By- Chandan Gupta Sir

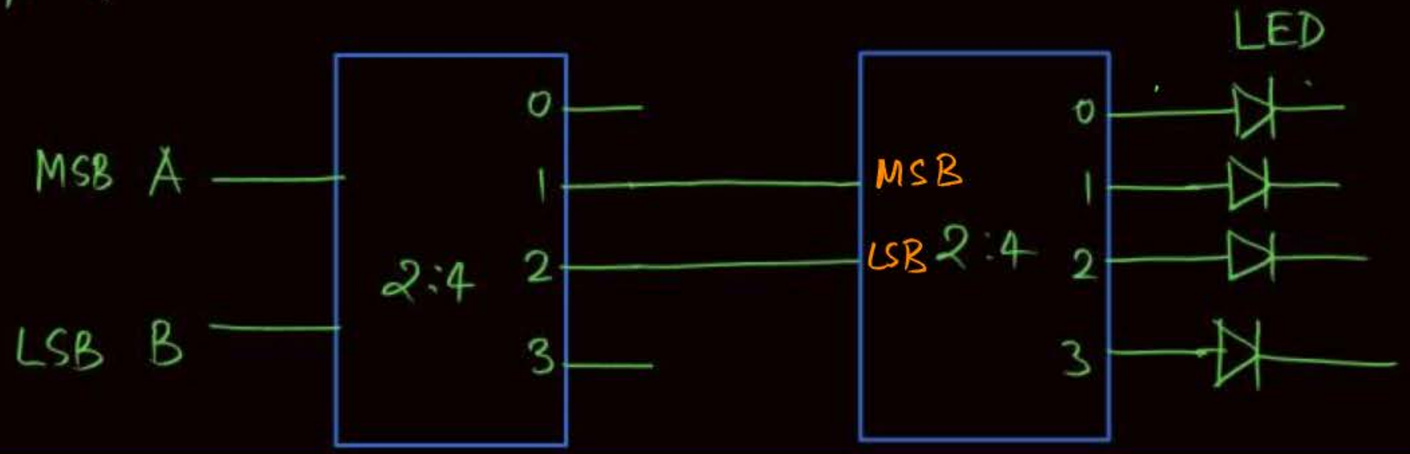


DeMUX & Decoder



Encoder

#Q.



00 → 0, 01 — 2, 10 — 1, 11 — 0

On I/P side we apply the different combination of A,B repeatedly then which no O/P LED will never glow ___3___.

Encoder

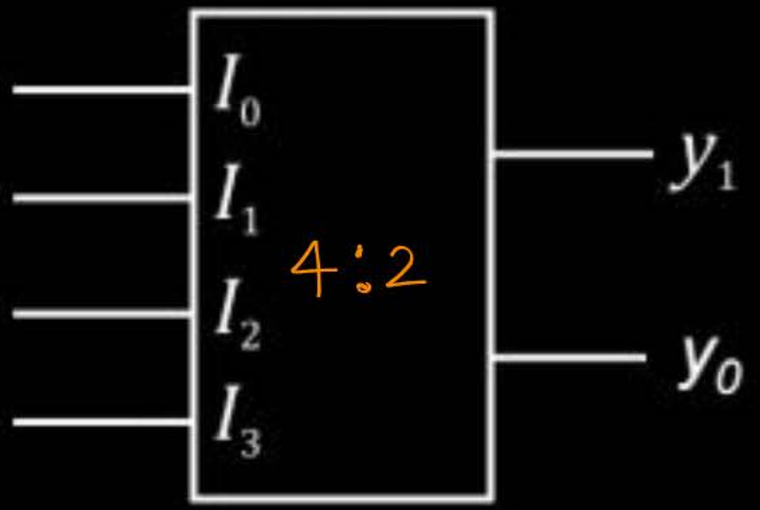
- What is encoder ?

- It is combinational ckt having many i/Ps and many o/Ps. It is basically used to convert other codes into binary.
  - Octal to binary [8:3] Encoder
  - Hexadecimal to binary [16:4] Encoder

- Order :→ $2^n : n$

- All the i/Ps lines can't be zero at the same time of Encoder.
  - All 0's can't be applied at the i/P of encoder.

- 4 : 2 Encoder :



$$y_1 = I_2 + I_3$$

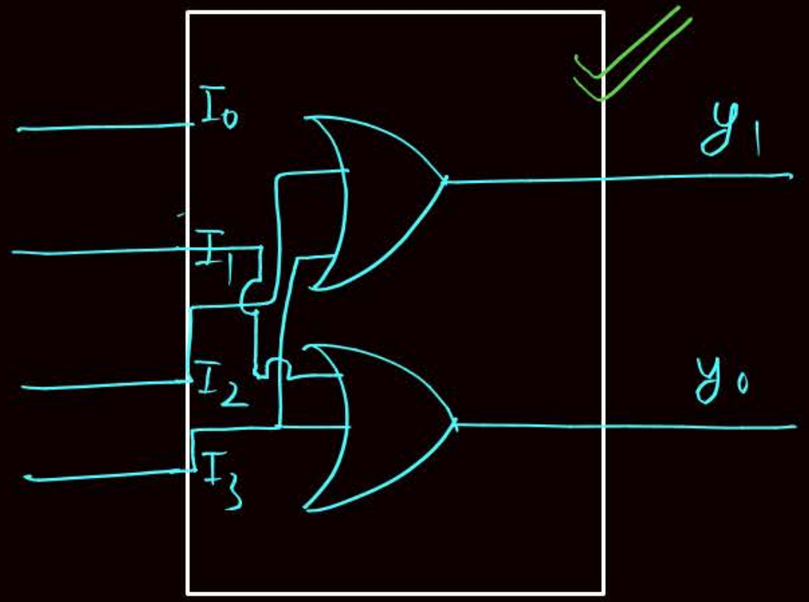
$$y_0 = I_1 + I_3$$

$y_1[I_0, I_1, I_2, I_3] = \Sigma(1,2) + d\Sigma(0,3,5,9,10,11,13,14)$

$y_0[I_0, I_1, I_2, I_3] = \Sigma(1,4) + d\Sigma(0,3,5,6,7,9,10,11,12,13)$

Truth Table :

	I_0	I_1	I_2	I_3	y_1	y_0
8	1	0	0	0	0	0
4	0	1	0	0	0	1
2	0	0	1	0	1	0
1	0	0	0	1	1	1



y_1

	$\bar{I}_2\bar{I}_3$	$\bar{I}_2 I_3$	$I_2 I_3$	$I_2\bar{I}_3$
$\bar{I}_0\bar{I}_1$	X	1	X	1
$\bar{I}_0 I_1$		X	X	X
$I_0 I_1$	X	X	X	X
$I_0\bar{I}_1$		X	X	X

I_2

I_3

$= I_2 + I_3$

y_0

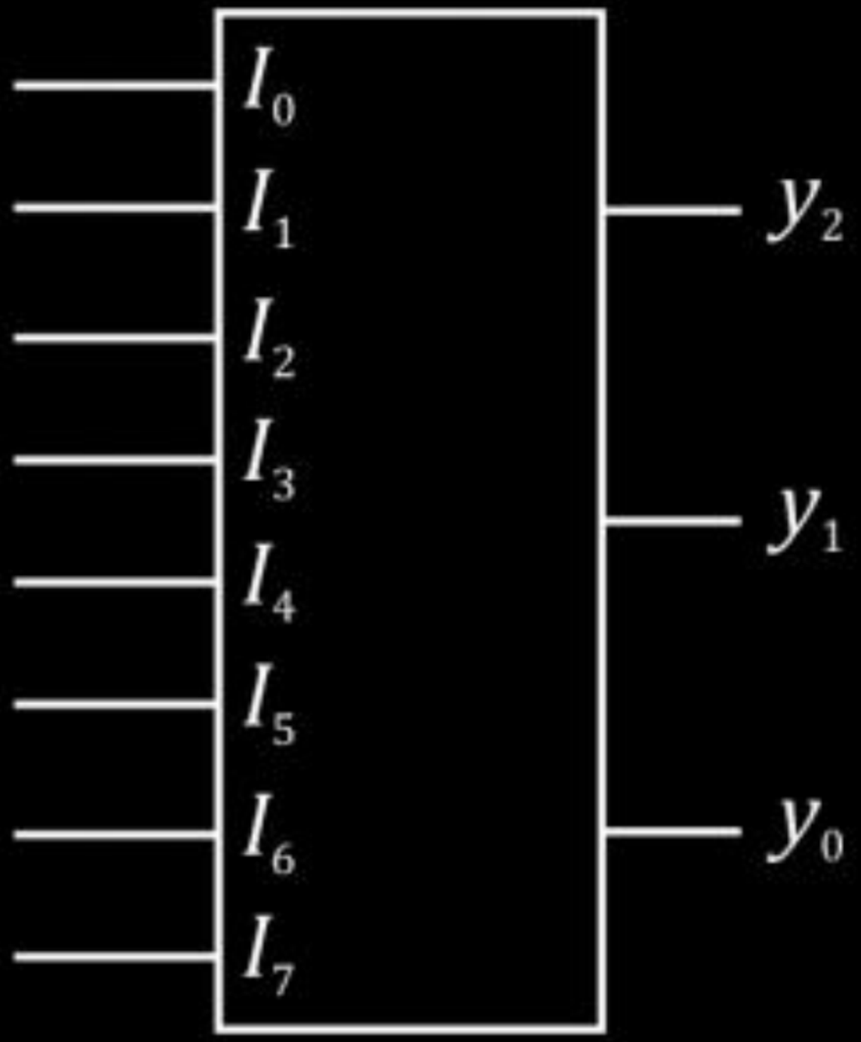
	$\bar{I}_2\bar{I}_3$	$\bar{I}_2 I_3$	$I_2 I_3$	$I_2\bar{I}_3$
$\bar{I}_0\bar{I}_1$	X	1	X	
$\bar{I}_0 I_1$	1	X	X	X
$I_0 I_1$	X	X	X	X
$I_0\bar{I}_1$		X	X	X

I_3

I_1

$= (I_1 + I_3)$

- 8 : 3 Encoder :



Truth Table :

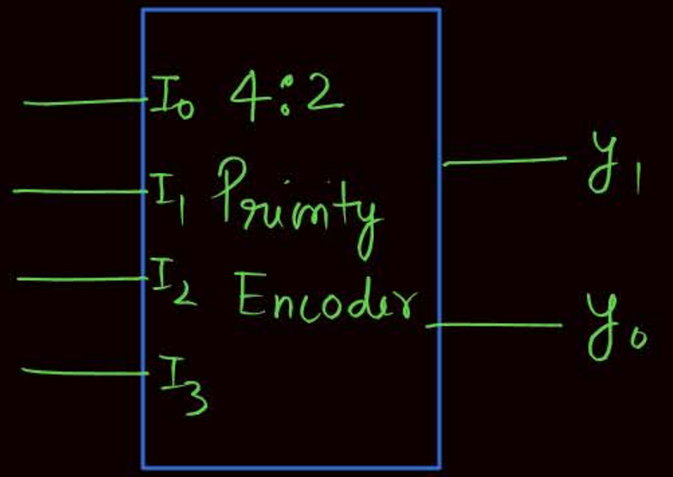
I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7	y_2 (W)	y_1 (Y)	y_0 (Z)
1	0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0
0	0	0	1	0	0	0	0	0	1	1
0	0	0	0	1	0	0	0	1	0	0
0	0	0	0	0	1	0	0	1	0	1
0	0	0	0	0	0	1	0	1	1	0
0	0	0	0	0	0	0	1	1	1	1

$$y_2 = I_4 + I_5 + I_6 + I_7$$

$$y_1 = I_2 + I_3 + I_6 + I_7$$

$$y_0 = I_1 + I_3 + I_5 + I_7$$

Priority Encoder

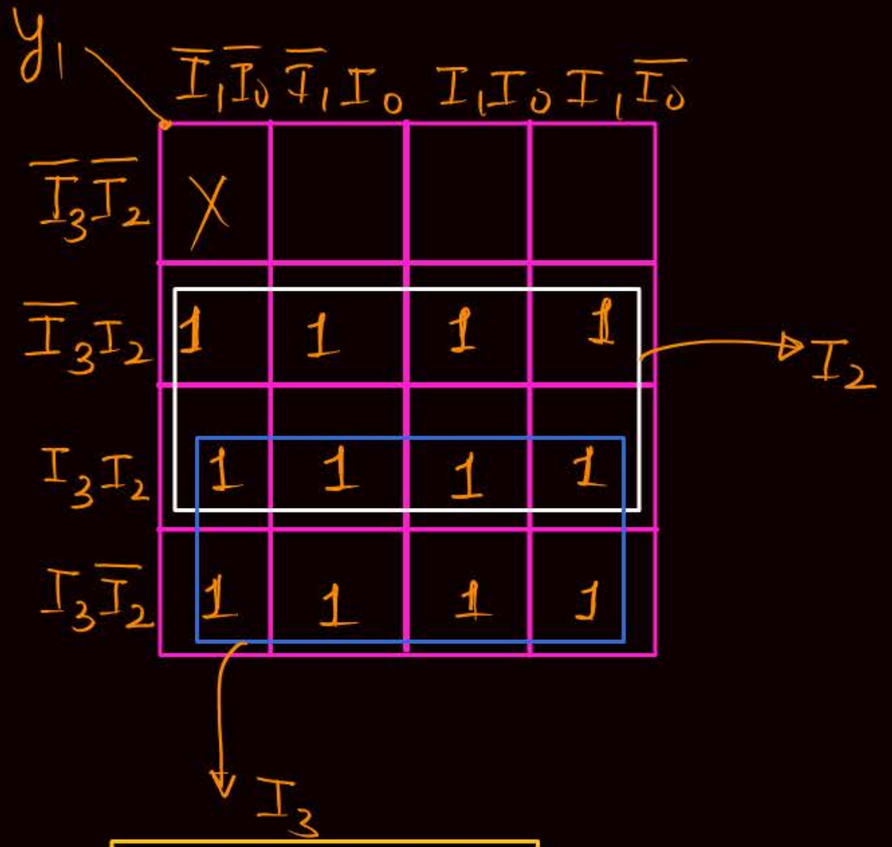


$$I_3 > I_2 > I_1 > I_0$$

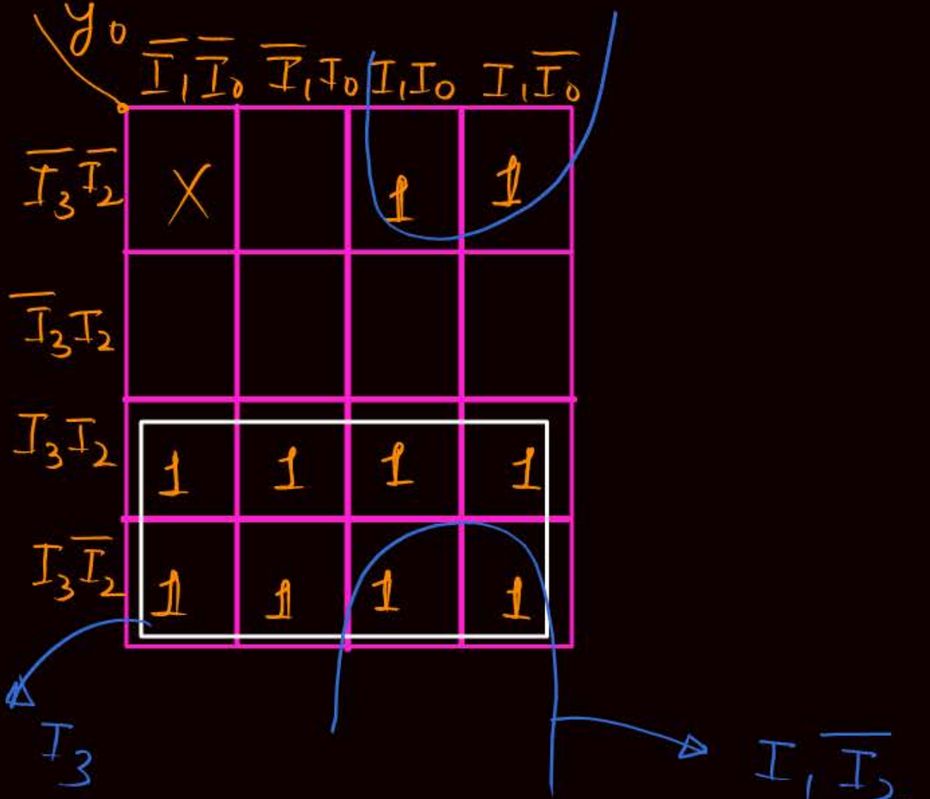
$$y_1[I_3, I_2, I_1, I_0] = \sum 4, 5, 6, 7 \ldots 15) + d \Sigma 0$$

$$y_0[I_3, I_2, I_1, I_0] = \Sigma 2, 3, 8, 9 \ldots 15) + d \Sigma 0$$

I_3	I_2	I_1	I_0	y_1	y_0
0	0	0	0	X	X
0	0	0	1	0	0
0	0	1	0	0	1
0	0	1	1	0	1
0	1	0	0	1	0
0	1	0	1	1	0
0	1	1	0	1	0
0	1	1	1	1	0
1	0	0	0	1	1
1	0	0	1	1	1
1	0	1	0	1	1
1	0	1	1	1	1
1	1	0	0	1	1
1	1	0	1	1	1
1	1	1	0	1	1
1	1	1	1	1	1



$$y_1 = (I_2 + I_3)$$

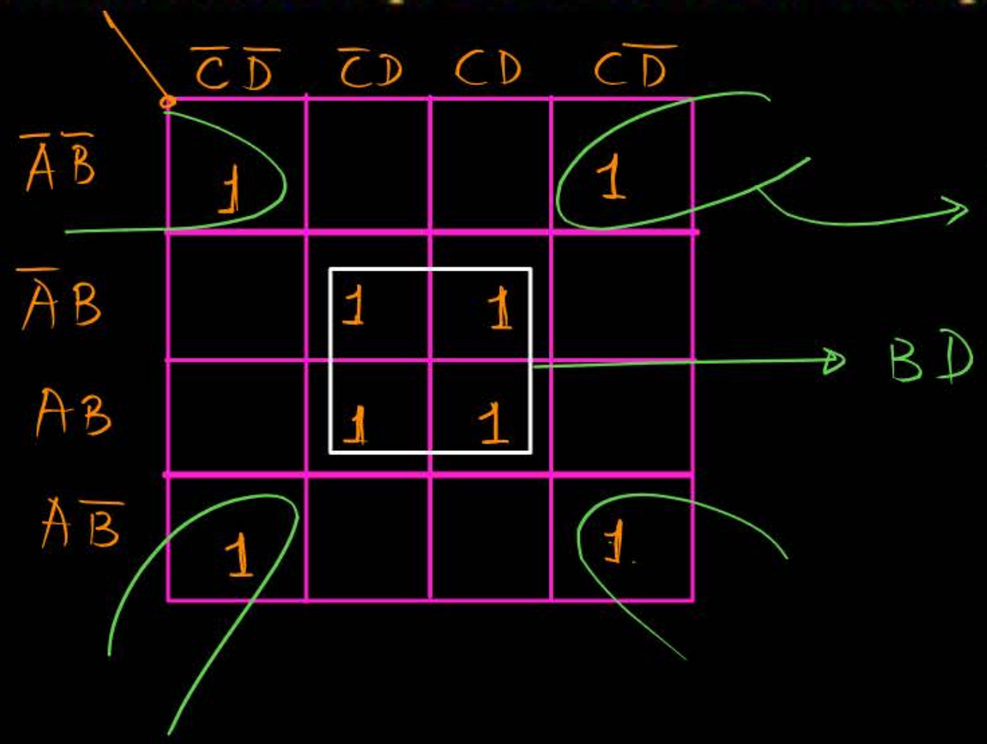


$$y_0 = I_3 + I_1\bar{I}_2$$

[NAT]

What is the minimum number of 2-input NOR gates required to implement a 4-variable function expressed in sum-of-minterms form as f = Σ(0, 2, 5, 7, 8, 10, 13, 15)? Assume that all the inputs and their complements are available. 3

[GATE-2019-CS: 2M]



$$= \bar{B}\bar{D} + BD$$

$$= (\bar{B}+D)(B+\bar{D})$$

$$f = (P+D)(B+Q)$$

$$\bar{f} = \overline{(P+D)} + \overline{B+Q}$$

$\overline{(P+D)}$ → NOR W

$\overline{B+Q}$ → NOR Z

$$\bar{f} = W + Z$$

$$f = \overline{W+Z}$$ → 1 NOR

[MCQ]

What is the minimal form of the karnaugh map shown below? Assume that X denotes a don't care term

[GATE-2012-CS: M]

cd \ ab	00	01	11	10
00	1	×	×	1
01	×			1
11				
10	1			×

$\bar{C}\,\bar{b}$

$\bar{b}\,\bar{d}$

A. $\bar{b}\bar{d}$

B. $\bar{b}\bar{d} + \bar{b}\bar{c}$

C. $\bar{b}\bar{d} + a\bar{b}\bar{c}d$

D. $\bar{b}\bar{d} + \bar{b}\bar{c} + \bar{c}\bar{d}$

[MCQ]

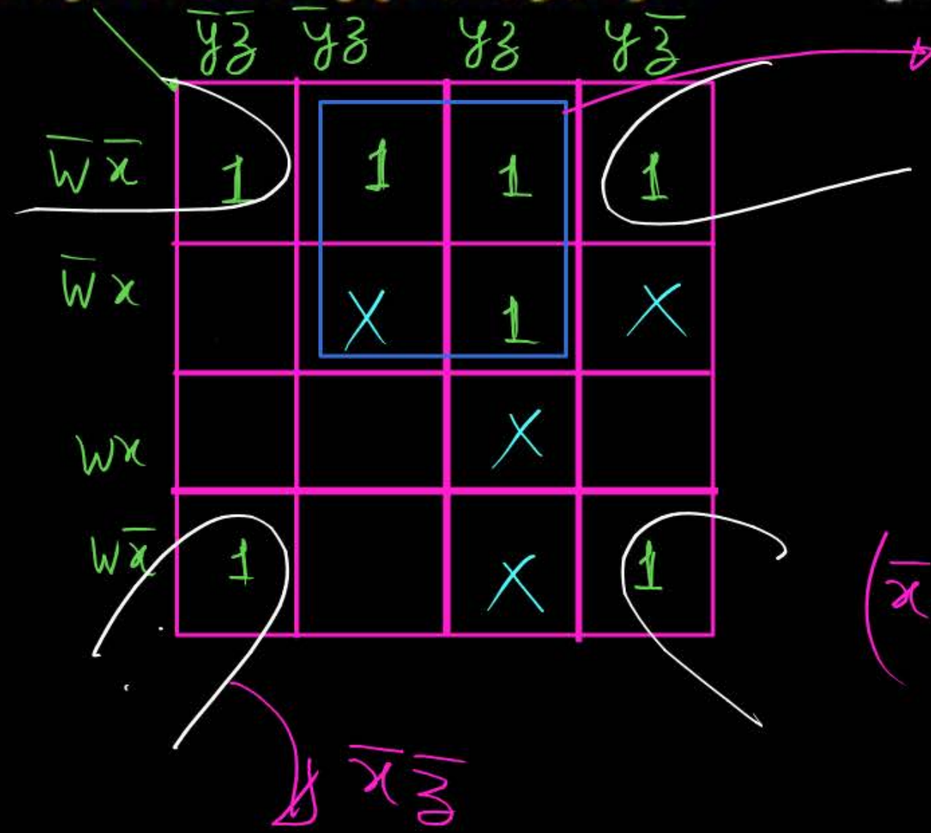
Given $f(w, x, y, z) = \Sigma m(0, 1, 2, 3, 7, 8, 10) + \Sigma d(5, 6, 11, 15)$, where d represents the don't-care condition in Karnaugh maps. Which of the following is a minimum product-of-sums (POS) form of $f(w, x, y, z)$? [GATE-2017-CS: 1M]

A. $f = (\bar{w} + \bar{z})(\bar{x} + z)$

B. $f = (\bar{w} + z)(x + z)$

C. $f = (w + z)(\bar{x} + z)$

D. $f = (w + \bar{z})(\bar{x} + z)$

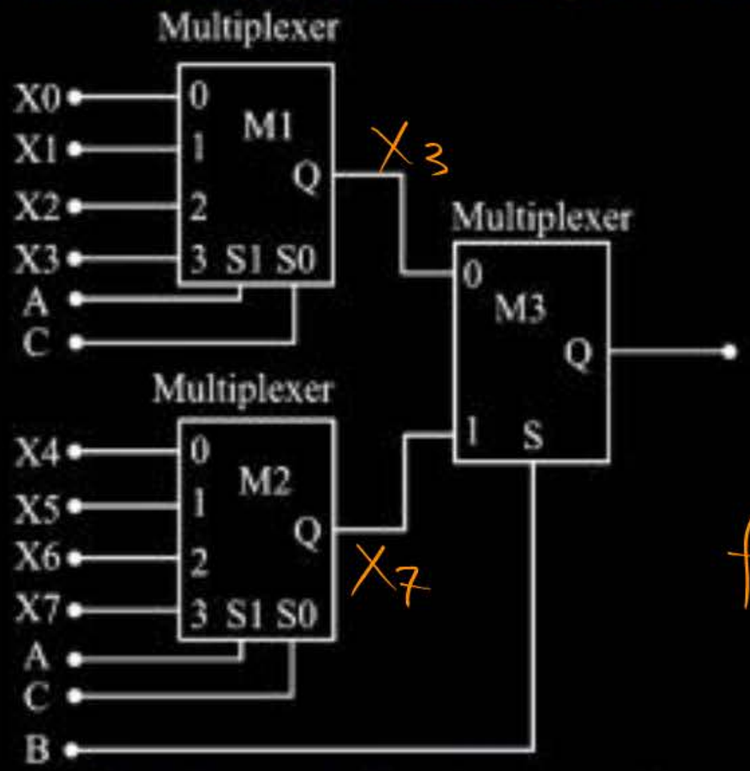


$\bar{w}z + \bar{x}\bar{z}$

$(\bar{x} + z)(\bar{w} + \bar{z})$

[MCQ]

A Boolean digital circuit is composed using two 4-input multiplexers (M1 and M2) and one 2-input multiplexer (M3) as shown in the figure. X0-X7 are the inputs of the multiplexers M1 and M2 and could be connected to either 0 or 1. The select lines of the multiplexers are connected to Boolean variables A, B and C as shown.



Which one of the following set of values of (X0, X1, X2, X3, X4, X5, X6, X7) will realise the Boolean function?

[GATE-2023-CS: 2M]

A (1, 1, 0, 0, 1, 1, 1, 0)

B (1, 1, 0, 0, 1, 1, 0, 1)

C (1, 1, 0, 1, 1, 1, 0, 0)

D (0, 0, 1, 1, 0, 1, 1, 1)



2 Minute Summary

Encoder.

Thank you

GW

Soldiers !

